

PRD.md

Product Requirements Document

Product Name

SynChain

Vision

Build an intelligent decision-support platform that allows organizations to simulate, evaluate, and optimize supply chain decisions before executing them.

Problem

Supply chain teams often rely on spreadsheets, fragmented systems, and historical reports.

These methods are reactive rather than predictive.

As a result:

- Excess inventory
- Stock shortages
- Poor warehouse utilization
- Delayed response to disruptions

can occur.

Solution

A Multi-Agent AI Supply Chain Digital Twin that:

- Creates a virtual model of supply chain operations
- Runs specialized agents
- Generates recommendations
- Explains decisions

Target Users

- Supply Chain Managers
- Inventory Planners
- Operations Teams
- Logistics Coordinators

Current Features

- Supply chain simulation
- Demand analysis
- Inventory recommendation
- Logistics recommendation
- Risk assessment

Planned Features

- Scenario simulation
- Multi-agent orchestration
- Digital twin state tracking

Future Features

- External intelligence integration
- News monitoring
- Supplier disruption analysis
- Commodity price tracking

Success Metrics

- Faster decision making
- Reduced stock imbalances
- Improved inventory utilization
- Increased operational visibility